SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

Saruhasini S
192521418RD

CO1_AT3_OUTPUT PREDICTION

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QUESTIONS

- **Q1** Reverse a list — 10 marks
- **Q2** Reverse a string — 10 marks
- **Q3** Searching — 10 marks
- **Q4** Searching — 10 marks
- **Q5** Array — 10 marks
- **Q6** Merging — 10 marks

Q1 Reverse a list

10 marks

Write a C program that takes an array as input and returns it reversed.

Your Code

```c
#include<stdio.h>
int main()
{
    int n,a[100],i;
    scanf("%d",&n);
    for(i=0; i<n; i++)
      scanf("%d",&a[i]);
    for(i=n-1; i >=0; i--)
      printf("%d ",a[i]);
    return 0;
}
```

Input

```
5
10 20 30 40 50
```

Output

```
50 40 30 20 10
```

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- **Q2** Reverse a string — 10 marks
- **Q3** Searching — 10 marks
- **Q4** Searching — 10 marks
- **Q5** Array — 10 marks
- **Q6** Merging — 10 marks
- **Q7**

Q2 Reverse a string — 10 marks

Write a C program for palindrom

Your Code

```c
#include <stdio.h>
int main() {
 int n, t, r, rev = 0;
 scanf("%d", &n);
 t=n;
while(n>0){
    r = n % 10;
    rev = rev * 10 + r;
    n /= 10;
}
    if(t == rev)
        printf("palindrome");
    else
        printf("not a palindrome");
    return 0;
}
```

Input

121

Output

```
palindrome
```

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QUESTIONS

- Q1 Reverse a list 10 marks
- Q2 Reverse a string 10 marks
- Q3 Searching 10 marks
- Q4 Searching 10 marks
- Q5 Array 10 marks
- Q6 Merging 10 marks
- Q7 Largest and Smallest element 10 marks
- Q8

Q3 Searching

10 marks

Write a C program for Searching an Element in a Sorted Array

Your Code

```
#include<stdio.h>
int main()
{
    int n, a[100],key;
    int low = 0,high,mid,i;
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&key);
    high=n - 1;
    while(low <= high)
    {
        mid = (low + high) /2;
        if(a[mid] == key)
        {
            printf("Found at position %d", mid + 1);
            return 0;
        }
        else if(a[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
```

Input

```
5
10 20 30 40 50
40
```

Output

```
Found at position 4
```

QUESTIONS

Q1
Reverse a list
10 marks

Q2
Reverse a string
10 marks

Q3
Searching
10 marks

Q4
Searching
10 marks

Q5
Array
10 marks

Q6
Merging
10 marks

Q7
Largest and Smallest element
10 marks

Q8
Second Largest Element
10 marks

Q4 **Searching**

10 marks

Write a C program for Count the Occurrences of an Element

Your Code

```
int main()
{
    int n, a[100], key;
    int low, high, mid, i;
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&key);
    low=0;
    high=n-1;
    while(low<=high)
    {
        mid=(low+high)/2;
        if(a[mid]==key)
        {
            printf("%d",mid+1);
            return 0;
        }
        else if(a[mid]<key)
            low=mid+1;
        else
            high=mid-1;
    }
    printf("Not found");
    return 0;
```

Input

```
10 20 30 40 50
40
```

Output

```
Not found
```

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QUESTIONS

Q1
Reverse a list
10 marks

Q2
Reverse a string
10 marks

Q3
Searching
10 marks

Q4
Searching
10 marks

Q5
Array
10 marks

Q6
Merging
10 marks

Q7
Largest and Smallest element
10 marks

Q8
Second Largest Element

Q5 **Array**

10 marks

Write a program for binary search

Your Code

```c
#include<stdio.h>
int main()
{
    int n, a[100], key;
    int i, count = 0;
    scanf("%d", &n);
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &key);
    for(i = 0; i < n; i++)
    {
        if(a[i] == key)
            count++;
    }
    printf("%d",count);
    return 0;
}
```

Input

```
5
10 20 30 20 20
20
```

Output

```
3
```

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QUESTIONS

Q1
Reverse a list
10 marks

Q2
Reverse a string
10 marks

Q3
Searching
10 marks

Q4
Searching
10 marks

Q5
Array
10 marks

Q6
Merging
10 marks

Q7
Largest and Smallest element
10 marks

Q8
Second Largest Element
10 marks

Q6 Merging

10 marks

Write a C program for Merge Two Arrays

Your Code

```c
#include<stdio.h>
int main()
{
    int n1, n2, a[100], b[100], c[200];
    int i, j = 0;
    scanf("%d", &n1);
    for(i = 0; i < n1; i++)
        scanf("%d", &a[i]);
    scanf("%d", &n2);
    for(i = 0; i < n2; i++)
        scanf("%d",b[i]);
    for(i = 0; i < n1; i++)
        c[j++] = a[i];
    for(i = 0; i < n2; i++)
        c[j++] = b[1];
    for(i = 0; i < j; i++)
       printf("%d ", c[i]);
             return 0;
 }
```

Input

```
3
10 20 30
3
40 50 60
```

Output

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QUESTIONS

Q1
Reverse a list
10 marks

Q2
Reverse a string
10 marks

Q3
Searching
10 marks

Q4
Searching
10 marks

Q5
Array
10 marks

Q6
Merging
10 marks

Q7
Largest and Smallest element
10 marks

Q8
Second Largest Element

Q7 **Largest and Smallest element** 10 marks

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```
#include <stdio.h>
int main()
{
    int a[100], n, i, max, min;
    scanf("%d", &n);
    for(i = 0; i < n; i++) scanf("%d", &a[i]);
    max = min = a[0];
    for(i = 1; i<n; i++)
    {
        if(a[i] > max) max = a[i];
        if(a[i] < min) min = a[i];
    }
        printf("Largest = %d\nsmallest = %d", max, min);
        return 0;
        }
```

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Input

```
5
10 25 3 40 15
```

Output

```
Largest = 40
smallest = 3
```

QUESTIONS

Q1
Reverse a list
10 marks

Q2
Reverse a string
10 marks

Q3
Searching
10 marks

Q4
Searching
10 marks

Q5
Array
10 marks

Q6
Merging
10 marks

Q7
Largest and Smallest element
10 marks

Q8
Second Largest Element
10 marks

Q8 Second Largest Element

10 marks

Write a C program for Find the Second Largest Element

Your Code

```c
#include<stdio.h>
int main()
{
    int a[10],n,i,max,s;
    scanf("%d",&n);
    for(i=0;i<n;i++)
        max=s=a[0];
    for(i=1;i<n;i++)
        if(a[i]>max){s=max;max=a[i];}
    else if(a[i]>s&&a[i]!=max) s=a[i];
    printf("%d",s);
    return 0;
}
```

Input

```
5
10 20 30 40 50
```

Output

```
0
```

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- **Q2** Reverse a string — 10 marks
- **Q3** Searching — 10 marks
- **Q4** Searching — 10 marks
- **Q5** Array — 10 marks
- **Q6** Merging — 10 marks
- **Q7** Largest and Smallest element — 10 marks

Q9 Occurrence of an elemnet — 10 marks

Write a C program for Find the First and Last Occurrence of an Element

Your Code

```c
#include <stdio.h>
int main() {
    int a[20], n, x, i, f=-1, l=-1;
    scanf("%d",&n);
    for(i=0;i<n;i++) scanf("%d",&a[i]);
        scanf("%d",&x);
        for(i=0;i<n;i++){
            if(a[i]==x){
                if(f==-1) f=i;
            }
        }
        printf("First=%d\nLast=%d", f,l);
        return 0;
}
```

Input

```
5
2 4 2 6 2
2
```

Output

```
First=0
Last=-1
```

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Q1 Reverse a list 10 marks

Q2 Reverse a string 10 marks

Q3 Searching 10 marks

Q4 Searching 10 marks

Q5 Array 10 marks

Q6 Merging 10 marks

Q7 Largest and Smallest element 10 marks

Q10 Missing Element

10 marks

Write a C program for Find Missing Element in an Array

Your Code

```c
#include <stdio.h>
int main()
{
    int a[20], n, i, sum = 0;
    scanf("%d", &n);
    for(i = 0; i < n - 1; i++){
        scanf("%d", &a[i]);
        sum += a[i];
    }
    printf("%d", n * (n + 1) / 2 - sum);
    return 0;
}
```

Input

```
5
1 2 3 4
```

Output

```
5
```

Submitted